CSE 6179 – Digital Forensics

Hands-on Lab # 2 Data Acquisition (RAM Imaging Techniques) Spring 2026

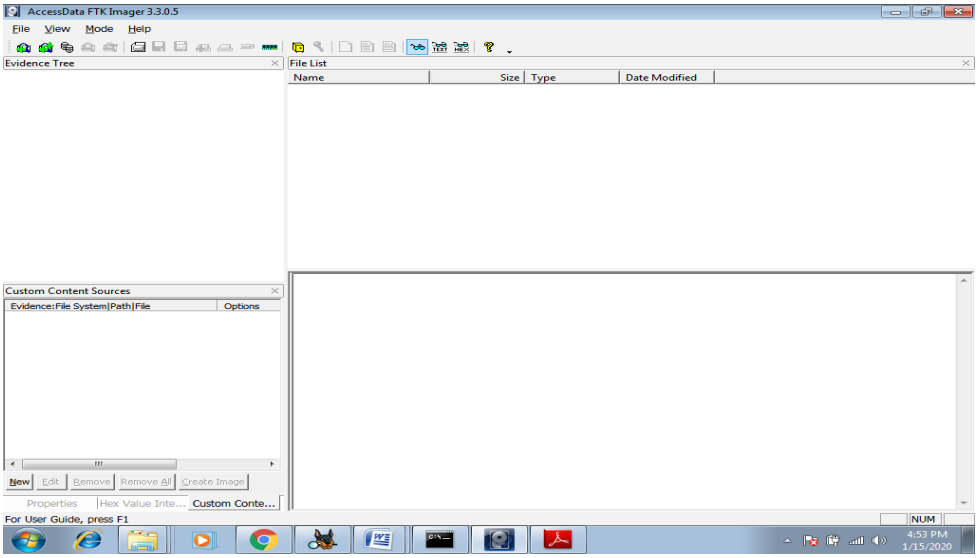
2.1 Theoretical Background

The first pro-active step in any digital forensic investigation is that of acquisition. The inherent problem with digital media is that it is readily modified; even just by accessing files. For this reason analysts obtain a "bit copy" of the media using specialist tools which stop modification occurring. Working from a copy is one of the fundamental steps to making a forensic investigation auditable and acceptable to a court. Another fundamental part of the process is the ability to verify the accuracy of the evidence produced; acquisition and verification are key concepts in preparing digital media for investigation.

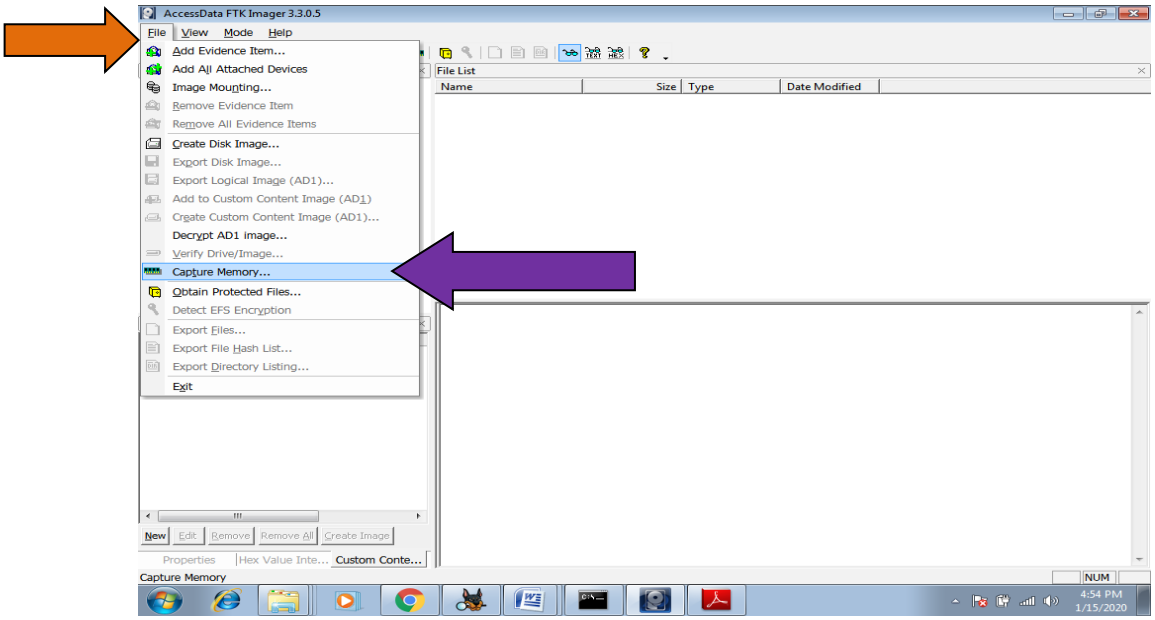
2.2 Instructions

How to image of the volatile memory/physical memory (RAM) by using "FTK Imager" Software

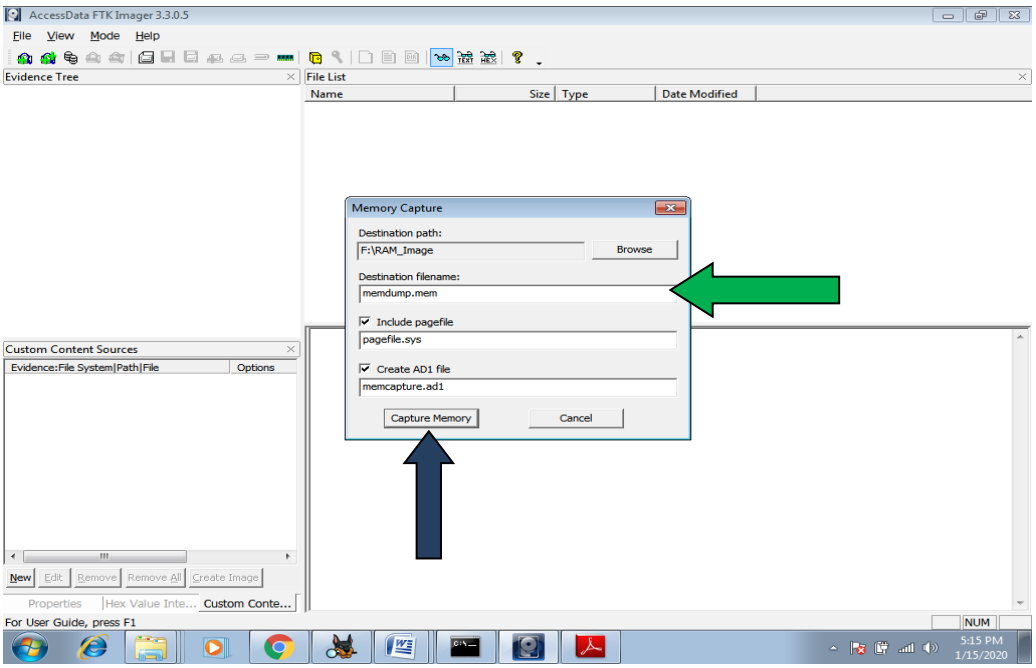
1. Double/Single clicks the "AccessData FTK Imager" icon from your PC.



2. Click "File" from the menu bar then go to "Capture Memory" and open it.



3. Browse to the destination path where available space is and then click "Capture Memory" Button.

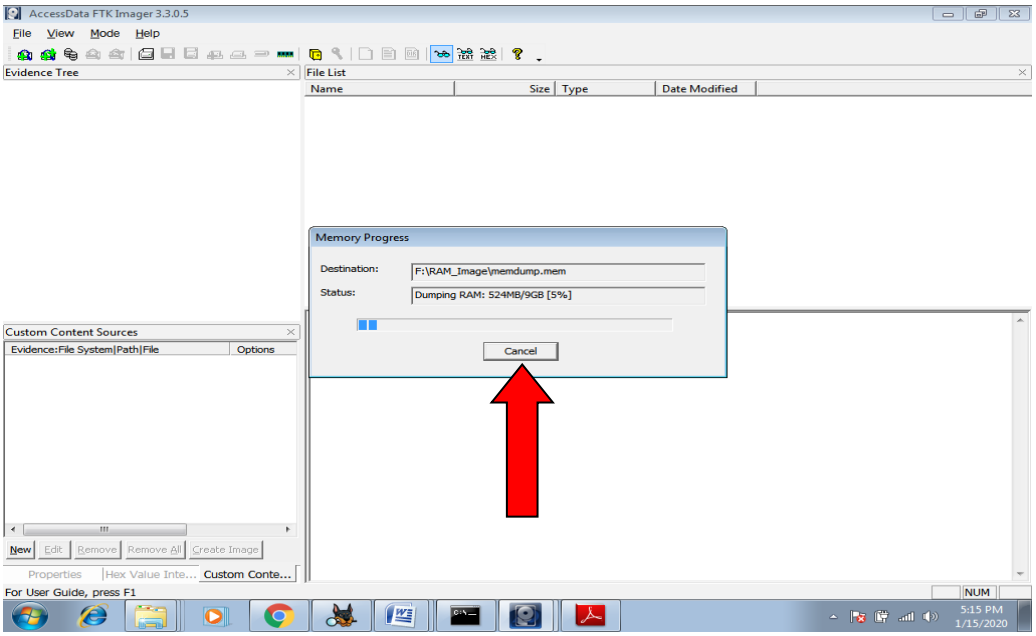


Pagefile: The pagefile (pagefile.sys) is used in Windows operating systems as volatile memory due to limitation of physical random-access memory (RAM). It is located under the "C" partition ready to use as volatile memory when the existing RAM capacity is exceeded. So this file can have quite a bit of valuable data when considering the volatile memory. Therefore, it is recommended to capture and collect this file in the acquisition.

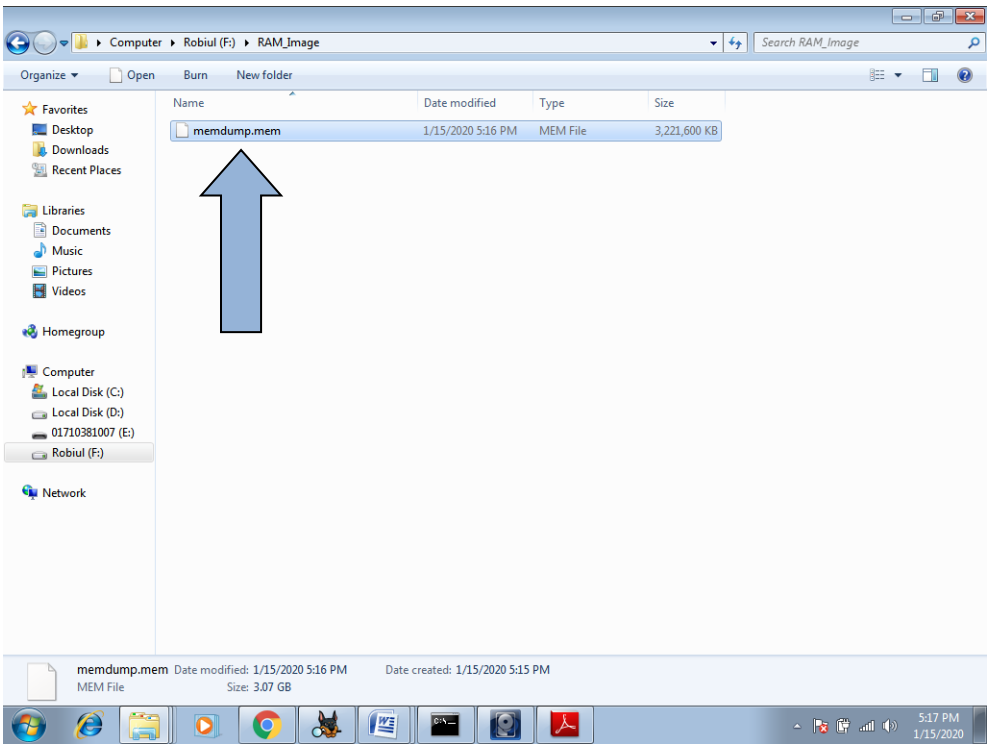
AD1 file: AD1 is the FTK imager image file. The investigator has the option to create an AD1 file for later use.

NOTE: Once the acquisition has completed, the destination folder will have the acquired memory with the file extension of ".mem".

4. Here is shown that the imaging is running; don't click the "**Cancel**" Button.



5. After imaging completion, we will get "memdump.mem" image file in our destination Folder.

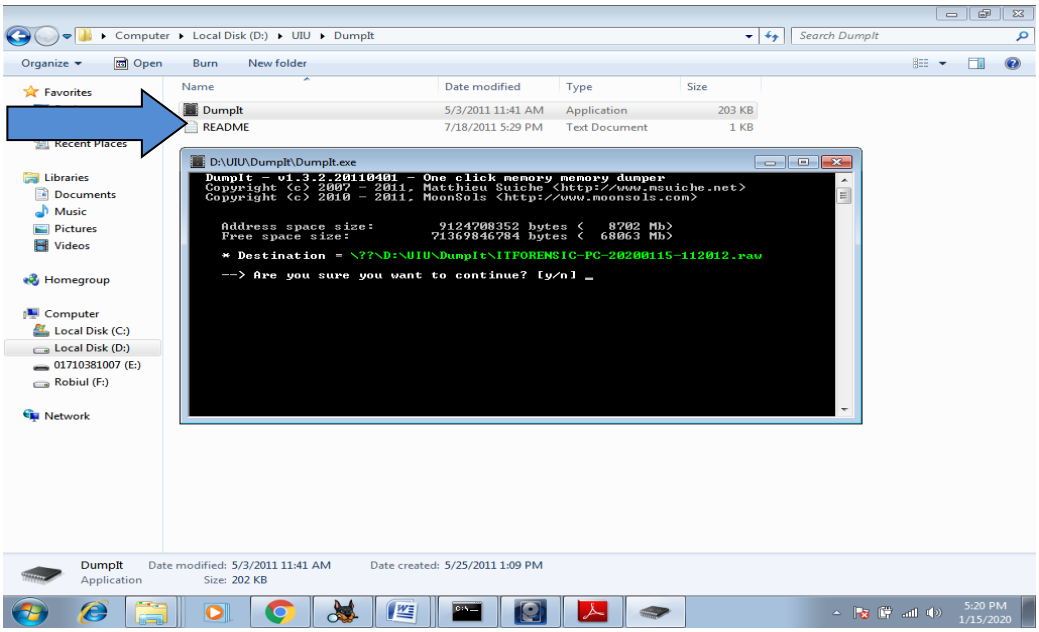


How to image of the volatile memory/physical memory (RAM) by using "DumpIt" Software

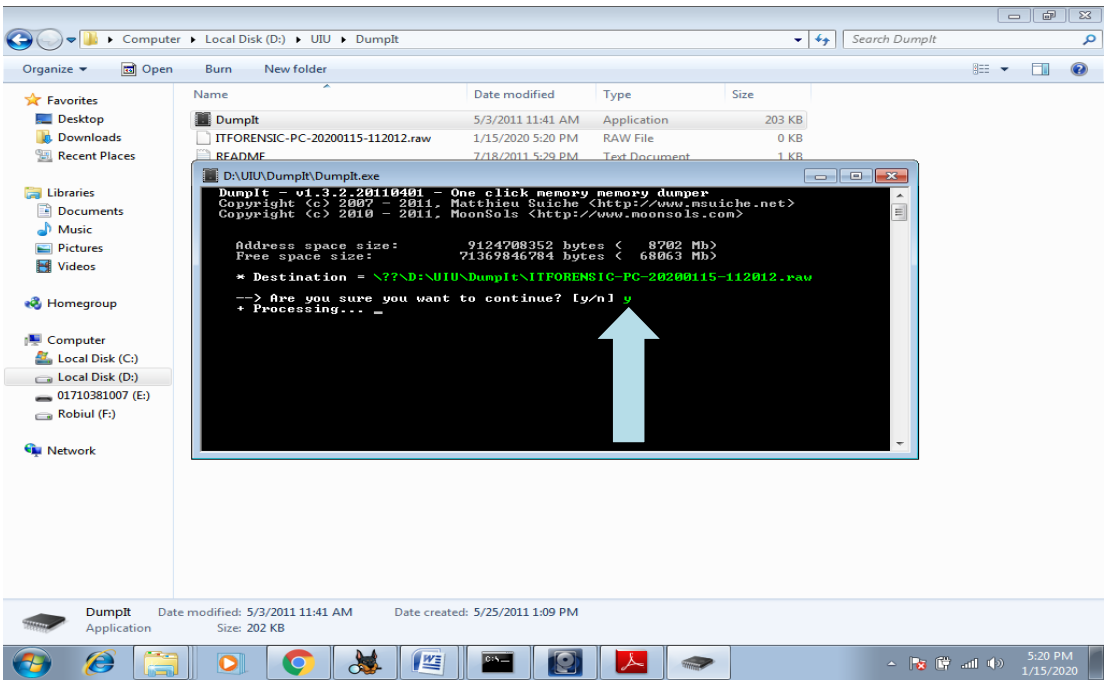
To collect any evidence from Victim PC, one of the first steps is to capture the victim's RAM. This can be possible using dumpit tool. This utility is used to generate a physical memory dump of Windows machines. It works with both x86 (32-bits) and x64 (64-bits) machines. The raw memory dump is generated in the current directory, only a confirmation question is prompted before starting. Perfect to deploy the executable on USB keys, for quick incident responses needs.

Run Dumpit.exe file the raw memory dump will be generated and save to the same directory

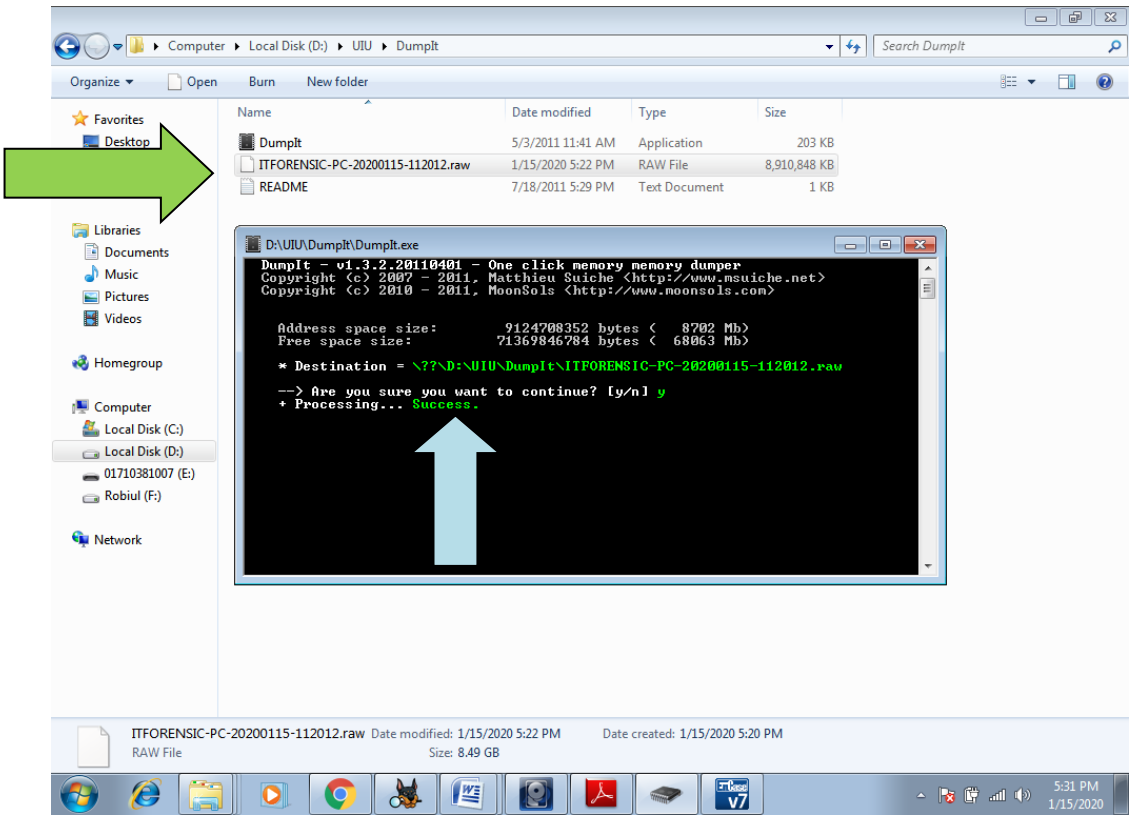
1. Double/Single clicks the "DumpIt" icon from your PC.



2. Type character "y" by using your keyboard and waiting for completion image.



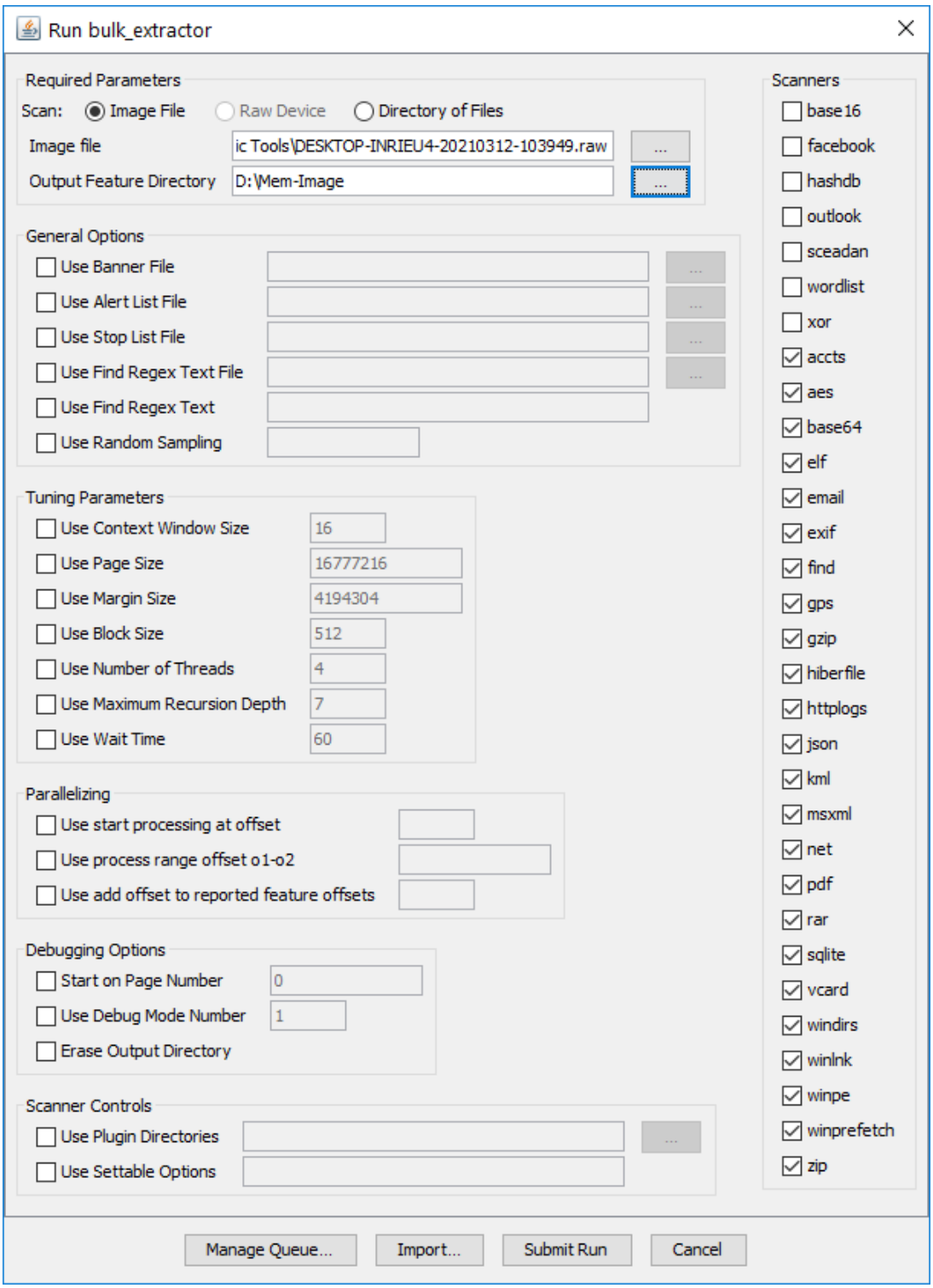
3. When you will see "Success" on the black screen, it means that the image is done.



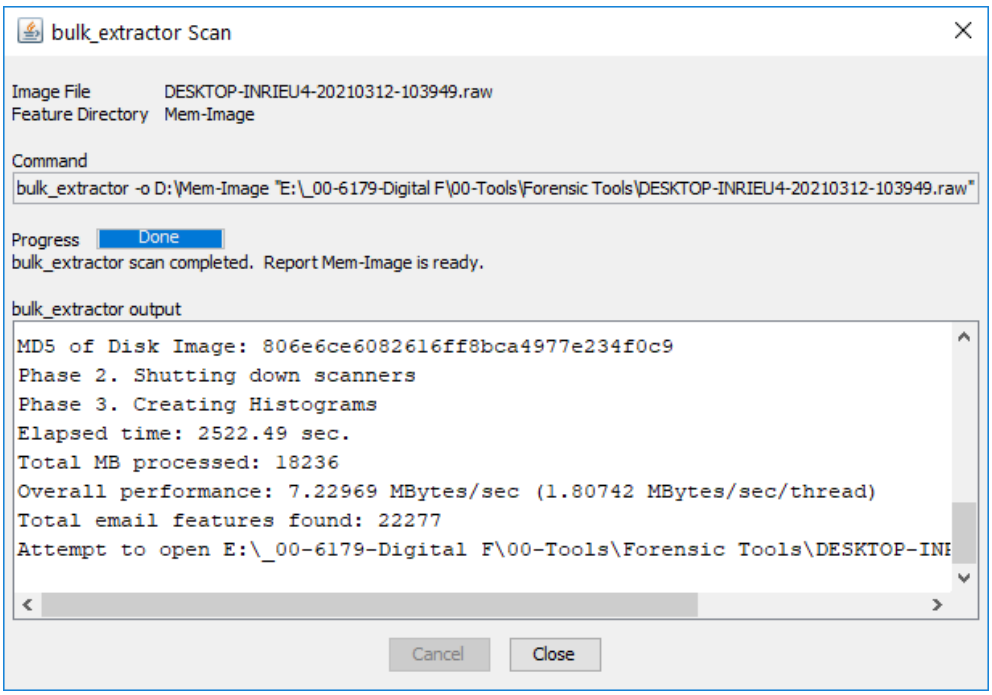
How to Collect Email Evidence in Victim PC

1. Download **bulk extractor viewer** from shared Google drive and install it in your PC. Now open **bulk extractor viewer** and click on to **Tools – Run Bulk Extractor**.

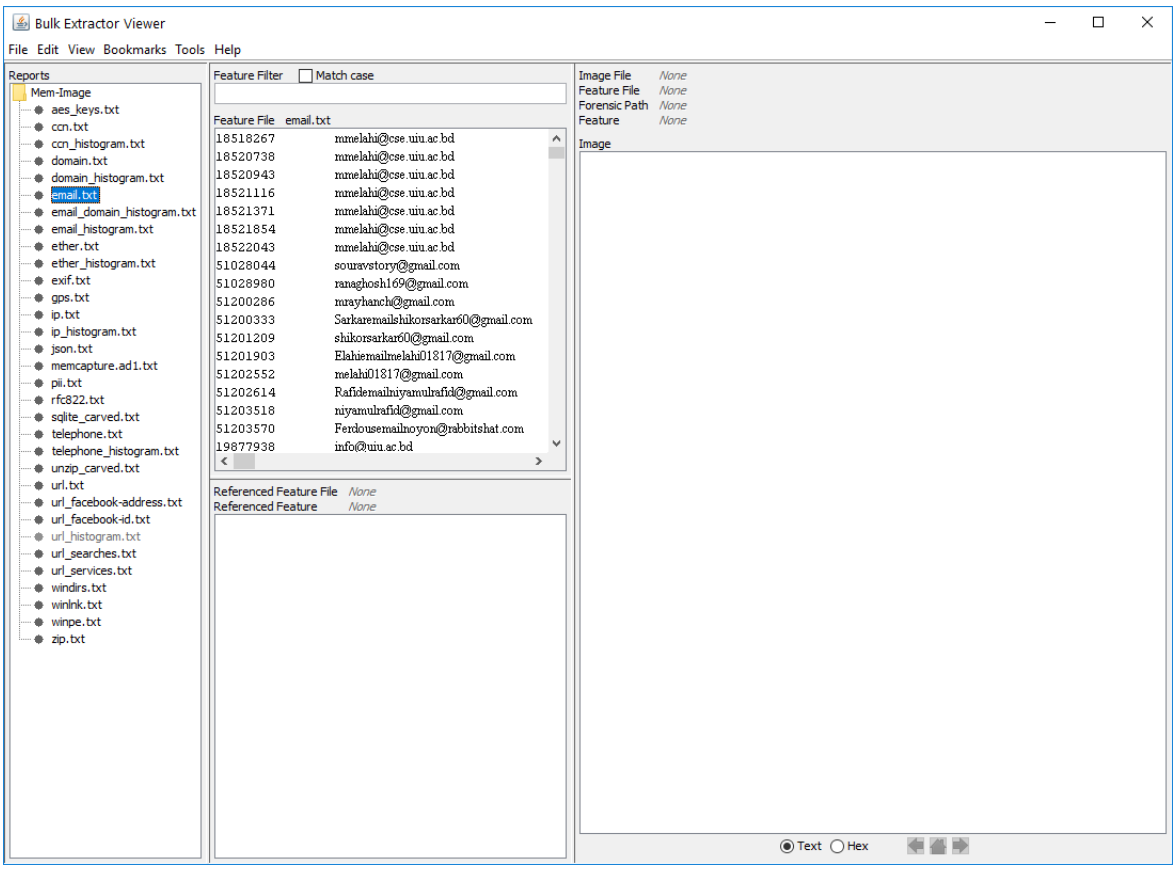
2. Now, select the **dump it image file** and select an **output folder** for the report and click on "**Submit Run**" as seen below:



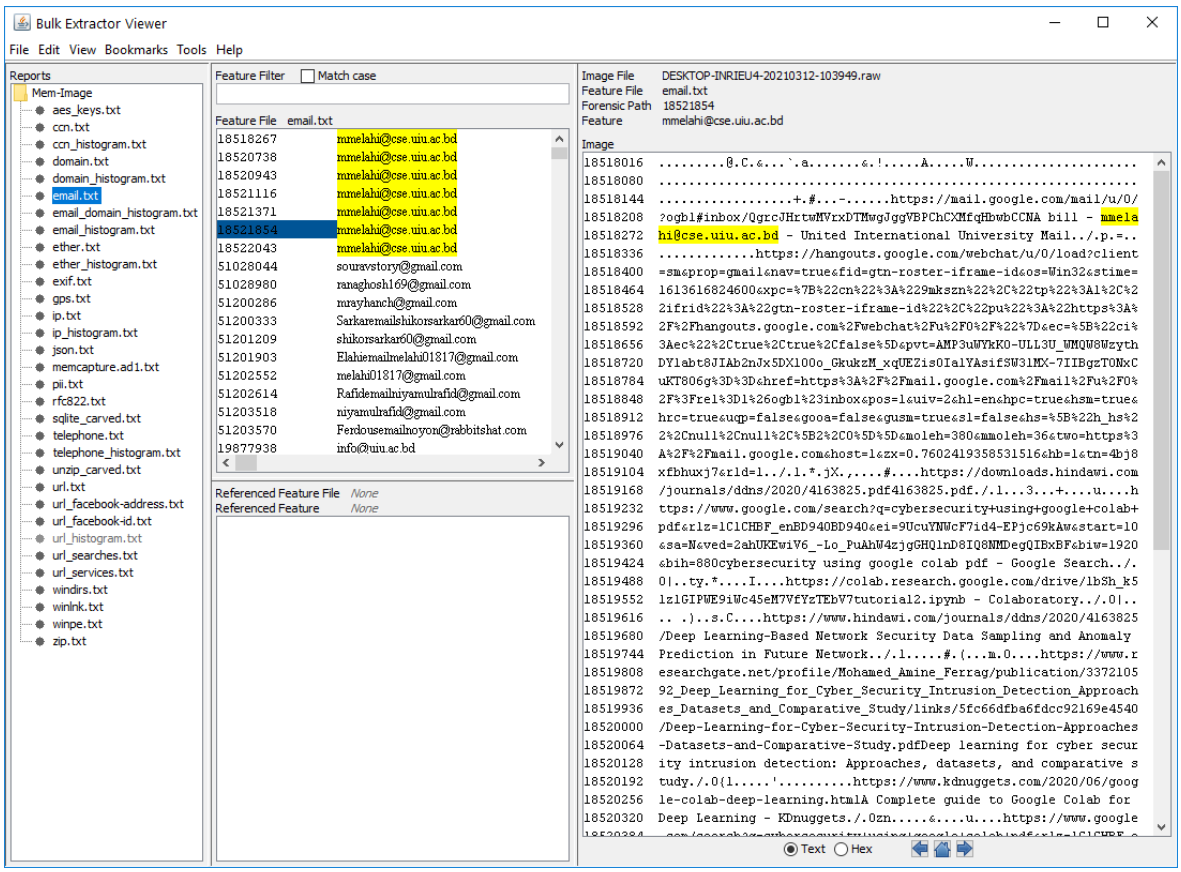
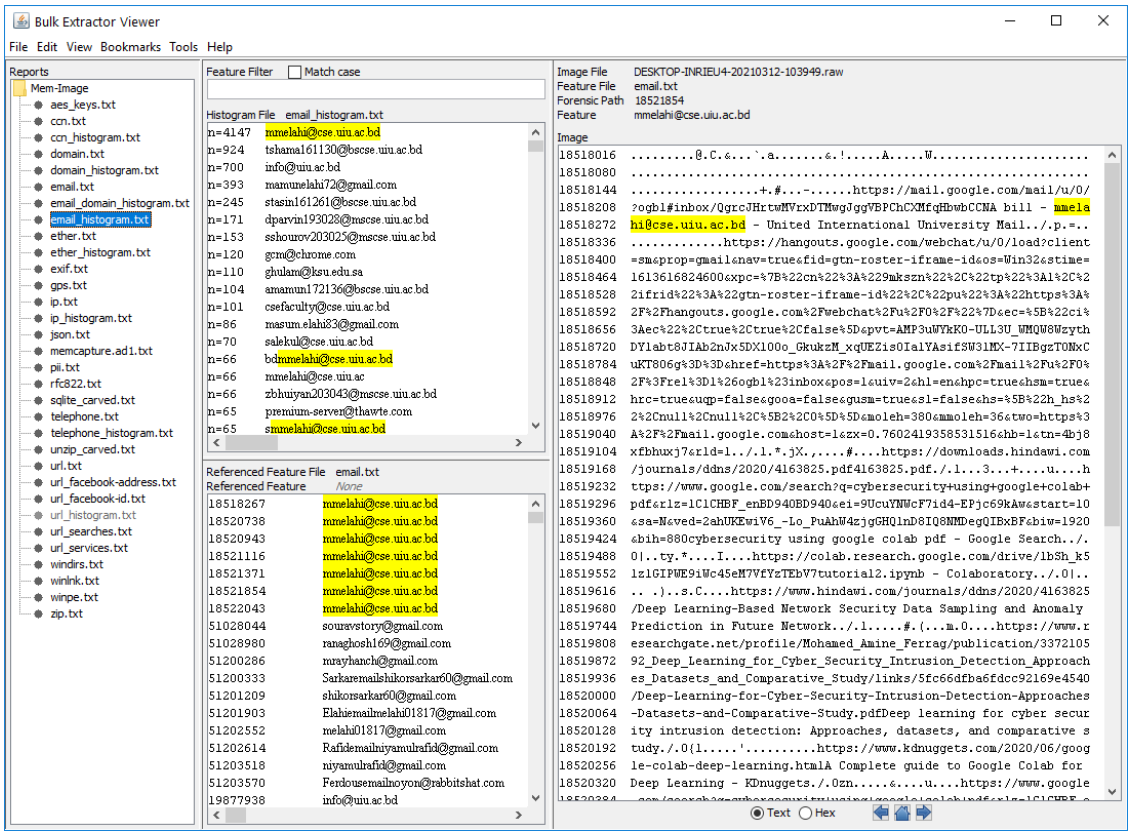
3. When the run is complete, click **"Close".**



3. Go back to Bulk Extractor Viewer to the main Bulk Extractor Viewer window to view the results of the run. The "**Reports**" window on the left will now show the **newly created report**. Clicking once on this report name will expand the report and show all of the files that have been created. Clicking on one of the files will bring that file up in the "**Feature File**" window in the middle of the screen as shown in the following figure. For example, in order to **investigate the victim saved information of Email ID** Click on **"email.txt"** as seen below:.



4. While viewing the feature file, the user can **select a feature** to view with it"s **full context** in the feature file. In this case, selecting "**mmelahi@cse.uiu.ac.bd"**, shows the feature in context within the **feature file on the right hand side** of the window as shown in the following Figure:



5. The user can **view histogram files** in the Bulk Extractor Viewer. Clicking on the file, "**email_histogram.txt**" in the Reports window on the left hand side will bring up the **contents of the histogram file** in the middle window. It will also display the **referenced feature file** in the window below the histogram file. In this case, the referenced feature file is **"email.txt"**. Clicking on a feature in the histogram, in this example mmelahi@cse.uiu.ac.bd, will display the feature in context as found within the feature file on the right hand side of the screen as shown in the following Figure: